

Shape Optimization of the Triangular Vortex Flowmeter

Numerical Simulation and Optimization using LBM

PROJECT AIM

Use an LBM-based flow simulation to study vortex shedding from a triangular bluff body, calculate useful flow quantities, fit the numerical data, and determine an improved geometry using numerical optimization.

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FLOW → TRIANGULAR BLUFF BODY → VORTEX SHEDDING

The selected paper is the reference; our numerical pipeline is designed around it.

BASE PAPER

Shape Optimization of the Triangular Vortex Flowmeter Based on the LBM

Method

Sun et al., Symmetry (2025)

Paper's CFD model

- D3Q19 LBM
- Multiple-relaxation-time (MRT)
- LES turbulence treatment
- OpenLB framework

What we take from it

- Triangular vortex-flowmeter problem
- Vortex shedding as the key signal
- Geometry affects frequency and pressure loss
- Shape optimization as the main task

Our numerical contribution

- Systematic numerical-data analysis
- Interpolation + curve fitting
- Golden-section / Newton optimization
- Round-off and truncation error study

WHAT IS THE BLUFF BODY?

It is an intentional obstruction placed inside the pipe.

- It separates the incoming flow.
- The separated flow forms alternating vortices.
- The vortices create periodic pressure/force fluctuations.
- A sensor can detect this periodic signal.

No obstruction → no intentionally generated vortex signal.

WHY TRIANGULAR?

- The shape controls how the flow separates.
- Changing its dimensions changes vortex strength and regularity.
- Changing the shape also changes pressure loss.

Design variables

**h = characteristic height
 θ = selected geometric angle
other shape parameters if included**

DESIGN TRADE-OFF

Too weak → difficult signal detection

Too strong → unnecessary pressure loss

Target

stable, measurable vortex signal + acceptable pressure loss

This becomes the optimization objective.

OpenLB is used as the LBM simulation framework; we analyze its outputs with our numerical methods.

WHY LBM?

LBM is a CFD numerical method that evolves discrete particle distribution functions on a lattice.

Model chosen to follow the base paper

- D3Q19: 3 dimensions with 19 discrete velocity directions.
- MRT: different moments can relax at different rates.
- LES: represents unresolved turbulent effects with a sub-grid model.
- OpenLB: open-source LBM framework used by the paper.

We do NOT need to reproduce the paper's huge MRT transformation matrix in the proposal; OpenLB handles the solver implementation.

LBM EVOLUTION — CONCEPTUAL EQUATION

$$F_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) \\ = f_i(\mathbf{x}, t) + \text{collision term}$$

Meaning

:

- f_i = distribution associated with lattice direction i
- $\mathbf{e}_i$ = corresponding discrete velocity
- Δt = simulation time step
- collision term drives the distribution toward equilibrium

The proposal focuses on what enters the solver and what comes out; the full implementation is handled through OpenLB.

INPUTS

Geometry

- pipe dimensions
- triangle geometry (h , θ , etc.)
- generator position

Fluid / flow

- inlet velocity U
- density ρ
- kinematic viscosity ν
- boundary conditions

LBM SIMULATION

D3Q19 + MRT + LES

OpenLB

time stepping

collision → streaming → boundaries

**repeat until the required flow regime
/ periodic signal is obtained**

**store selected time histories and
flow fields**

OUTPUTS

- velocity field $u(x,y,z,t)$
- pressure-related field
- vorticity field / derived vorticity
- lift-force or pressure signal
- pressure drop ΔP
- vortex-shedding frequency f

**These become the input data for
our numerical analysis.**

VORTICITY FROM VELOCITY

LBM gives velocity components. We calculate the spatial derivatives numerically.

For the 2D visualization plane:

$$\omega = \partial v / \partial x - \partial u / \partial y$$

Central-difference approximation:

$$\partial v / \partial x \approx [v(x+\Delta x) - v(x-\Delta x)] / (2\Delta x)$$

$$\partial u / \partial y \approx [u(y+\Delta y) - u(y-\Delta y)] / (2\Delta y)$$

Output: vorticity map → vortex cores and vortex pattern.

STROUHAL NUMBER

First obtain the vortex-shedding frequency f from the periodic signal.

If T = time between successive peaks:

$$f = 1 / T$$

Then:

$$St = fD / U$$

where D = characteristic pipe/body dimension and U = reference flow velocity.

No FFT is required for the core proposal; peak-to-peak period is sufficient.

NUMERICAL INTEGRATION

If pressure-drop or another signal varies with time, compute its average numerically.

Average over a time interval:

$$\Delta \bar{P} = (1/T) \int_0 \Delta P(t) dt$$

Implement:

- Trapezoidal rule
- Simpson's rule if the sampling is appropriate

Output: average pressure loss used in the design objective.

CURVE FITTING

Run LBM for several geometry values:

$$x \rightarrow J(x)$$

- Fit a polynomial / response curve.
- Estimate fitting error.
- Use the fitted curve to search for the optimum.

Example:

$$J(x) \approx a_0 + a_1x + a_2x^2$$

The coefficients are obtained numerically.

LEAST-SQUARES + QR

For fitted data $y \approx Xa$, choose a to minimize:

$$\|Xa - y\|^2$$

Use QR decomposition to solve the least-squares problem stably.

Output: fitted response $\hat{J}(x)$.

INTERPOLATION

LBM gives J only at selected geometries. Interpolation estimates J between them.

Compare:

- Lagrange polynomial
- Newton polynomial
- Spline interpolation

Example:

$$(x_1, J_1), (x_2, J_2), \dots \rightarrow \hat{J}(x)$$

Compare interpolation error and smoothness.

GOLDEN-SECTION SEARCH

Primary optimizer for a one-dimensional geometry parameter.

Input : $\hat{J}(x), x \in [a, b]$

Repeatedly shrink the interval containing the optimum.

Output : $x^* = \text{best geometry parameter}$

No derivative is required.

NEWTON'S METHOD

Optimization can also be written as a root-finding problem:

$$J'(x^*) = 0$$

Newton update:

$$x_{n+1} = x_n - J'(x_n) / J''(x_n)$$

Use numerical derivatives from the fitted curve.

Compare the optimum and convergence with golden-section search.

ROUND-OFF ERROR

Error caused by finite computer precision.

- Very small numerical quantities may be rounded.
- Repeated time-stepping can accumulate floating-point effects.
- We monitor whether results change meaningfully with numerical precision/settings.

TRUNCATION ERROR

Occurs because continuous operations are replaced by finite numerical approximations.

Example from our project:

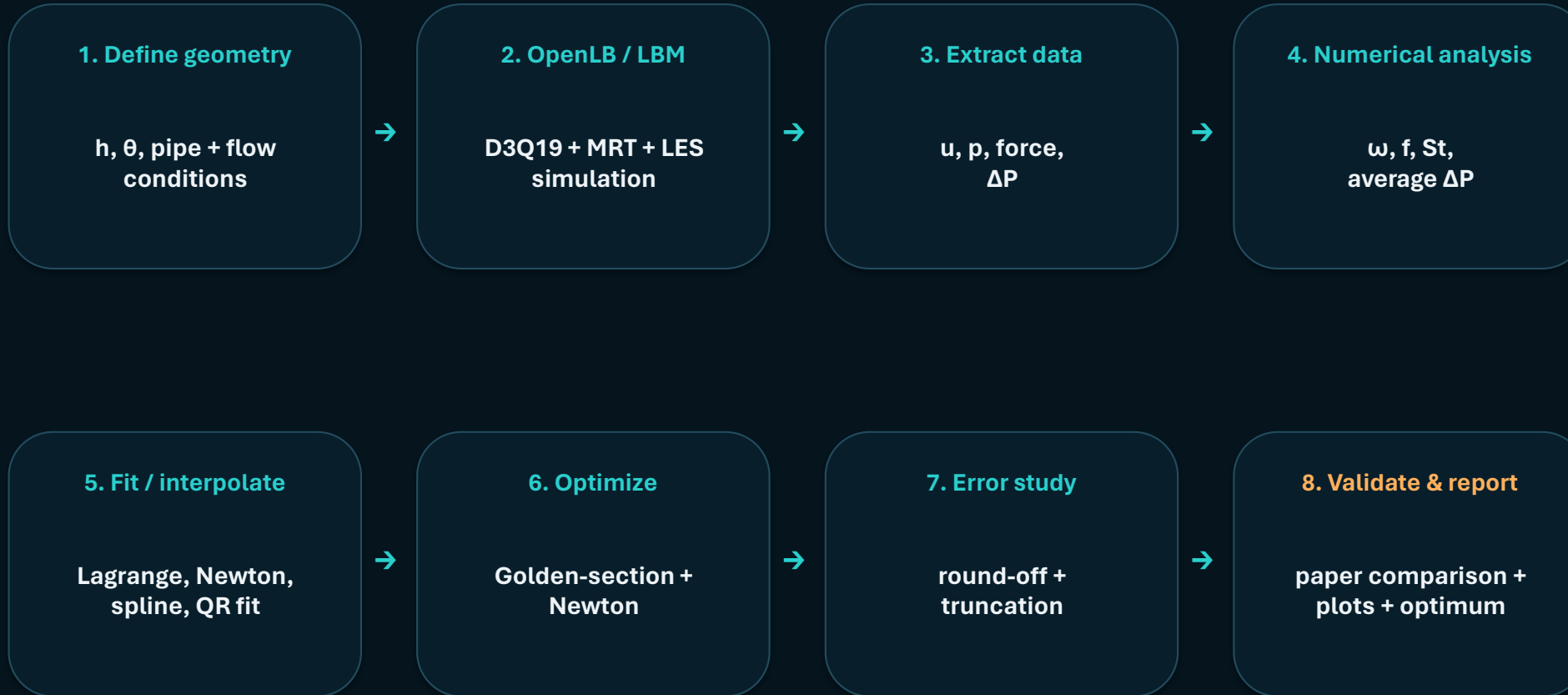
$$\partial v / \partial x \approx [v(x+\Delta x) - v(x-\Delta x)] / (2\Delta x)$$

The finite-difference formula is an approximation, so its result contains truncation error.

VALIDATION

- Compare selected quantities with the base paper.
- Check whether the same qualitative trend is obtained after geometry changes.

Complete Project Architecture and Expected Outputs



FINAL PROJECT OUTPUTS

Optimized geometry • vortex/pressure visualizations • St and ΔP • fitted response curve • numerical error report • comparison with base paper

WHAT WE WILL IMPLEMENT / PERFORM

- Use OpenLB for the LBM flow simulation based on the selected paper.
- Study the effect of triangular-generator geometry on vortex shedding and pressure loss.
- Calculate vortex frequency, Strouhal number and average pressure loss.
- Use Lagrange, Newton and spline interpolation.
- Use least-squares curve fitting with QR decomposition.
- Optimize the fitted response using golden-section search and Newton's method.
- Analyze round-off and truncation errors.
- Compare the obtained behavior with the base paper

EXPECTED RESULTS

- A working LBM simulation of the triangular vortex-flowmeter geometry.
- Flow-field and vortex visualizations.
- A numerical relationship between geometry and vortex/pressure metrics.
- A fitted/interpolated response curve for optimization.
- An estimated optimum geometry within the chosen design range.
- Quantified round-off/truncation-error discussion.
- A clear comparison between baseline and optimized designs.
- A comparison against the base paper's reported trends.

Feedback: We should have a clear idea on the topic